

PJM 5-Bus Battery Siting Problem

Mathematical Formulation

Square Peg — PQIC Phase 2

2026

Contents

1	Overview	1
2	Network and Sets	1
2.1	Sets and Indices	1
2.2	Network Data	2
3	Generator Data	2
3.1	Parameters	2
3.2	Generation Cost Function	2
4	Battery Data	2
4.1	Parameters	2
5	Decision Variables	3
5.1	Continuous Variables	3
5.2	Binary Variables (UC and Siting only)	3
6	Economic Dispatch Formulation	3
6.1	Objective	3
6.2	Constraints	3
7	Unit Commitment Formulation	4
7.1	Objective	4
7.2	Additional Constraints	4
8	Battery Siting Formulation	5
8.1	Structure	5
8.2	Search Space	5
8.3	Complexity	5
9	Summary Table	5

Overview

This document gives the complete mathematical formulation for the classical baseline of the battery siting problem on the PJM 5-bus network. Three optimizations are defined, each building on the last:

1. **Economic Dispatch (ED):** All generators are committed; find the least-cost output schedule including battery charge/discharge.
2. **Unit Commitment (UC):** Also decide which generators to turn on or off each hour. Binary decisions make this a mixed-integer problem.
3. **Battery Siting:** An outer loop over all placements of 2 batteries on the 5-bus grid. Runs UC for each of the $\binom{5}{2} = 10$ combinations and ranks by total cost.

The DC power flow approximation is used throughout. All costs are in USD.

Network and Sets

Sets and Indices

$\mathcal{N} = \{1, \dots, 5\}$	Buses (nodes)
$\mathcal{L} = \{1, \dots, 6\}$	Transmission lines
$\mathcal{G} = \{0, 1, 2\}$	Generators
$\mathcal{B} = \{0, 1\}$	Batteries
$\mathcal{T} = \{1, \dots, T\}$	Time steps (hours), $T \leq 24$

Network Data

Line	From–To buses	Flow limit $\bar{f}_\ell$ (MW)
1	1–2	500
2	1–4	unconstrained
3	1–5	unconstrained
4	2–3	unconstrained
5	3–4	unconstrained
6	4–5	350

The **Power Transfer Distribution Factor** (PTDF) matrix $H \in \mathbb{R}^{|\mathcal{L}| \times |\mathcal{N}|}$ maps nodal net injections to line flows under the DC approximation:

$$f_\ell = \sum_{n \in \mathcal{N}} H_{\ell n} p_n^{\text{net}}, \quad \ell \in \mathcal{L}.$$

Generator Data

Parameters

Each generator $g \in \mathcal{G}$ has output limits and a quadratic cost curve:

Unit	Bus	$\underline{p}_g$ (MW)	$\bar{p}_g$ (MW)	a_g (USD/MW ² h)	b_g (USD/MWh)	c_g (USD)
0	1	100	600	0.002	10	500
1	3	100	400	0.0025	8	300
2	5	50	200	0.005	6	100

Generation Cost Function

When generator g is operating at output $p_{g,t}$ in hour t , its cost is:

$$C_g(p_{g,t}) = a_g p_{g,t}^2 + b_g p_{g,t} + c_g.$$

The no-load cost c_g is incurred whenever the unit is *on*. In the ED formulation all units are always on; in UC it is multiplied by the binary commitment variable.

Battery Data

Parameters

Both batteries $b \in \mathcal{B}$ are identical Li-ion units sized for daily energy shifting (NREL ATB 2024):

Parameter	Value
Power rating $\bar{r}$	50 MW
Energy capacity $E^{\max}$	200 MWh
Minimum state of charge $E^{\min}$	0 MWh
Initial state of charge E^0	100 MWh
Round-trip efficiency η	0.85

Round-trip efficiency: charging the battery with 100 MWh of electricity stores only $\eta \cdot 100 = 85$ MWh of recoverable energy; the remaining 15% is dissipated as heat. The efficiency factor is applied to the charging side in the SOC update equation.

Decision Variables

Continuous Variables

Variable	Domain	Description
$p_{g,t}$	$\mathbb{R}_{\geq 0}$	Output of generator g at time t (MW)
$r_{b,t}^+$	$\mathbb{R}_{\geq 0}$	Charge power of battery b at time t (MW)
$r_{b,t}^-$	$\mathbb{R}_{\geq 0}$	Discharge power of battery b at time t (MW)
$e_{b,t}$	$\mathbb{R}_{\geq 0}$	State of charge of battery b at time t (MWh)

Binary Variables (UC and Siting only)

Variable	Domain	Description
$u_{g,t}$	$\{0, 1\}$	1 if generator g is on in hour t , else 0
$v_{g,t}$	$\{0, 1\}$	1 if generator g starts up in hour t , else 0
$z_{b,t}$	$\{0, 1\}$	1 if battery b is charging in hour t , else 0

Economic Dispatch Formulation

All generators are on for all t . No binary variables.

Objective

$$\min_{p, r^+, r^-, e} \sum_{t \in \mathcal{T}} \sum_{g \in \mathcal{G}} \left(a_g p_{g,t}^2 + b_g p_{g,t} + c_g \right)$$

Constraints

Power balance (at each bus n , each hour t):

$$\sum_{g: \text{bus}(g)=n} p_{g,t} + \sum_{b: \text{bus}(b)=n} (r_{b,t}^- - r_{b,t}^+) = d_{n,t}, \quad \forall n \in \mathcal{N}, t \in \mathcal{T}$$

where $d_{n,t}$ is the load at bus n in hour t .

Line flow limits:

$$-\bar{f}_\ell \leq \sum_n H_{\ell n} (p_{n,t}^{\text{gen}} + p_{n,t}^{\text{bat,net}} - d_{n,t}) \leq \bar{f}_\ell, \quad \forall \ell \in \mathcal{L}, t \in \mathcal{T}$$

Generator output limits:

$$\underline{p}_g \leq p_{g,t} \leq \bar{p}_g, \quad \forall g \in \mathcal{G}, t \in \mathcal{T}$$

Battery state-of-charge dynamics:

$$e_{b,t} = e_{b,t-1} + \eta r_{b,t}^+ - r_{b,t}^-, \quad \forall b \in \mathcal{B}, t \in \mathcal{T}$$

with $e_{b,0} = E_b^0$.

Battery SOC limits:

$$E^{\min} \leq e_{b,t} \leq E^{\max}, \quad \forall b \in \mathcal{B}, t \in \mathcal{T}$$

Battery power limits and no simultaneous charge/discharge:

$$r_{b,t}^+ \leq \bar{r} z_{b,t}, \quad r_{b,t}^- \leq \bar{r} (1 - z_{b,t}), \quad \forall b \in \mathcal{B}, t \in \mathcal{T}$$

Note: $z_{b,t}$ is binary in UC/Siting. In ED, if CVXPY does not enforce it as binary, it is treated as continuous in $[0, 1]$; the optimizer will naturally drive it to 0 or 1 because simultaneous charge and discharge is never economically optimal.

Unit Commitment Formulation

Extends ED by adding binary commitment decisions.

Objective

$$\min_{p, u, v, r^+, r^-, e} \sum_{t \in \mathcal{T}} \sum_{g \in \mathcal{G}} \left(a_g p_{g,t}^2 + b_g p_{g,t} + c_g u_{g,t} + S_g v_{g,t} \right)$$

where S_g is the startup cost of generator g .

Additional Constraints

Generator limits with commitment:

$$\underline{p}_g u_{g,t} \leq p_{g,t} \leq \bar{p}_g u_{g,t}, \quad \forall g \in \mathcal{G}, t \in \mathcal{T}$$

Startup indicator:

$$v_{g,t} \geq u_{g,t} - u_{g,t-1}, \quad \forall g \in \mathcal{G}, t \in \mathcal{T}$$

with $u_{g,0} = 0$. This forces $v_{g,t} = 1$ whenever a unit transitions from off to on; combined with $v_{g,t} \geq 0$ and the minimization, the optimizer sets $v_{g,t} = \max(0, u_{g,t} - u_{g,t-1})$.

All ED constraints (power balance, line flows, battery dynamics) apply unchanged. The problem class is **MIQP** (mixed-integer quadratic program), solved with SCIP.

Battery Siting Formulation

Structure

Siting is an outer loop over all placements of the two batteries on the 5-bus grid. It calls the UC solver as a black box for each placement:

$$\text{Siting: } \min_{\phi \in \Phi} C_{\text{UC}}^*(\phi)$$

where Φ is the set of all bus-pair assignments and $C_{\text{UC}}^*(\phi)$ is the optimal UC cost when batteries are placed at buses given by ϕ .

Search Space

With 5 buses and 2 batteries (unordered placement):

$$|\Phi| = \binom{5}{2} = 10$$

The 10 combinations are enumerated explicitly. For each $\phi \in \Phi$, the UC problem from Section 6 is solved with battery bus assignments fixed to ϕ . Results are ranked by $C_{\text{UC}}^*(\phi)$.

Complexity

Each UC solve is a MIQP with $3T$ binary variables ($T \leq 24$). Total wall-clock time is approximately $10 \times t_{\text{UC}}$, where $t_{\text{UC}} \approx 5\text{--}30$ seconds for $T = 24$, giving a siting runtime of roughly 1–5 minutes. Each UC solve is independent and can be parallelised.

Summary Table

Formulation	Binary vars	Continuous vars	Problem class	Solver	Est. runtime
ED	0	$9T$	QP	HiGHS	< 1 s
UC	$3T$	$9T$	MIQP	SCIP	5–30 s
Siting	$10 \times 3T$	$10 \times 9T$	MIQP $\times 10$	SCIP	1–5 min

Variable counts are for $T = 24$: $9T = 216$ continuous, $3T = 72$ binary per solve.